

BUSINESS ANALYSIS CASE STUDY 2

Campus Student Marketplace

TUT Soshanguve Campus — Student Commerce Research & Solution Design

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[STATUS: Complete]

1. Executive Summary

Students at TUT Soshanguve currently face high friction in campus commerce due to fragmented WhatsApp marketing and manual payment processes. Research revealed that while WhatsApp provides visibility, it causes message fatigue and lacks real-time inventory and scheduling tools. I propose "Student Online Mall," a centralized mobile application featuring real-time stock tracking, automated booking for services, and streamlined payment notifications. This solution is expected to increase seller revenue through wider campus visibility while significantly improving the buyer experience by removing transaction "awkwardness"

2. Problem Statement

Students at TUT Soshanguve Campus who run informal businesses currently rely on repeated WhatsApp group posts to market their products and services. This approach creates the following challenges:

- Constant repetitive posting causes message fatigue and frustration among group members.
- Businesses are only visible within specific residence sections and not across the wider campus.
- Customers have no way of knowing stock availability or service availability in real time.
- Students with serious business ambitions have no structured platform to grow beyond their immediate social circle.
- There is no centralized directory of campus businesses making it difficult for customers to find alternatives when one seller is out of stock.

3. Stakeholder Analysis

Two primary stakeholder groups have been identified for this case study:

3.1 Supply Side — Student Sellers

Stakeholders interviewed:

Stakeholder 1: Nail Technician

3.2 Demand Side — Student Buyers

As a student buyer, I decided to experience what it takes to buy from my fellow students, I realized that firstly I have to reply privately asking if the products are still available, then ask if they accept online payment, then after tell them what products I would like to buy, then sent my room number, on delivery I need to insert their number on my banking app to be able to transfer money which is an awkward waiting moment.

4. Research Findings

4.1 Seller Interviews

Key questions asked:

- Does constant posting in the group work for your business or is it frustrating?
- Would you prefer one place where students can find your business instead of posting daily?
- What is your biggest challenge with selling on campus?
- Would you use an app or website if it existed?

Findings:

Constant posting helps sellers as it act as a way of marketing their business, but also frustrates someone who is not interested in their business because they post at a central group where everyone gets constant notifications of something they are not interested in.

Currently they prefer posting because they are afraid of ending up neglecting school work and getting over appointments, they will also like a one place where their business will be recognized buy not only their resident mates but the whole campus which will allow the to work with different people and collaborate with student who offer the same service/product they offer.

The biggest challenge as service providers(nail tech) they cancel appointments because of upcoming test and exams, which also impact the trust, and causes conflicts and some student customer wants to look best when they write their tests/exams.

To resolve the constant posting frustrating , they will prefer a platform that only students who are interested to their service/product gets instant notification.

To resolve the challenge, they will like to use an app that they will limit customers they can service per day to avoid cancelling appointments that are out of their schedule and also get notification after a customer made an appointment so that they can prepare.

4.2 Buyer Interviews

Key questions asked:

- Do you know all the businesses operating in res?
- Have you ever wanted something but didn't know who sells it on campus?
- Would you use a platform where you can see what's available on campus?
- What would make you trust a campus seller more?

Findings:

- **Limited Awareness:** Most buyers only know of businesses within their specific block or floor; they often discover new sellers only by chance through scrolling past old WhatsApp messages.
- **Desired Visibility:** Buyers expressed a need for a "search" or "filter" function to find specific items (e.g., "earphones" or "braids") without waiting for a seller to post.
- **Trust & Reliability:** Trust would be significantly improved if the platform featured a rating or review system and a way to verify that a student is currently "on-duty" or has stock before walking to their room.
- **Payment Friction:** Buyers highlighted the "Awkward Waiting Moment" during EFT transfers as a major deterrent; they wish for a one-click payment or a digital wallet system.

4.3 Common Themes Identified

- **Information Asymmetry:** Both parties suffer from a lack of real-time data regarding availability and scheduling.
- **Communication Overload:** Sellers are overwhelmed by repetitive PMs, while buyers are overwhelmed by irrelevant group notifications.
- **Scheduling Conflicts:** A shared pain point where academic commitments (tests/exams) disrupt business operations, leading to canceled appointments and lost trust.
- **Payment Inefficiency:** The manual process of adding beneficiaries and showing proofs of payment is the highest friction point in the entire transaction.

AS-IS Process: Student Purchase Journey (WhatsApp/Manual)

Step	Actor	Action	Bottleneck
1	Buyer	Sees post in group and sends Private Message (PM)	Manual Effort: High friction to start.
2	Buyer	Asks about stock availability	Information Gap: No real-time stock view.
3	Seller	Confirms availability (eventually)	Time Lag: Seller might be in class.
4	Buyer	Asks for payment methods (Cash/EFT)	Inconvenience: Repetitive communication.
5	Buyer	Sends order details and delivery location (Room #)	Privacy: Sharing room numbers over text.
6	Seller	Delivers item to Buyer's location	Logistics: Manual coordination required.
7	Buyer	Adds Seller as beneficiary on banking app	CRITICAL PAIN: The "Awkward Waiting Moment."
8	Buyer	Performs transfer and shows proof	Security: Seller must trust the "screen swipe."

5. Requirements

5.1 Functional Requirements

FR1: Profile Management: Sellers must be able to create a business profile and upload a product/service catalog.

FR2: Real-Time Inventory: The system must allow sellers to update stock levels and toggle "Open/Closed" status.

FR3: Booking Engine: For service providers (e.g., Nail Techs), the app must support an appointment calendar with automated notifications.

FR4: Search & Filter: Buyers must be able to search for products by category (Food, Tech, Beauty).

FR5: Integrated Payments: Support for secure digital payment confirmation to eliminate the manual "beneficiary" setup.

FR6: Social Engagement: Ability for buyers to follow sellers and "like" posts to maintain a community feel.

5.2 Non Functional Requirements

NFR1: Performance: The app must be "data-lite" to ensure students with limited data bundles can access it easily.

NFR2: Security: Student data and room numbers must be encrypted and only visible during an active transaction.

NFR3: Usability: The interface must be mobile-first, as the majority of TUT students shop via smartphone.

NFR4: Availability: The platform should be accessible 24/7, even if individual shops are closed.

6. Proposed Solution

Solution Name: Student Online Mall

Solution Type: Application

customer side:

- it displays recent posts from the sellers, pictures or reels
- user can like, comment and follow to keep the social media feeling alive
- user can navigate to mall section where it displays products and services categorized
- on products side , user can see number of available items(drinks, snacks .etc.), user can select method of payment (cash on delivery or online transfer)
- on services side, user can make appointments, make payments and see weekly schedule of the service provider.

Seller side:

- seller can post their products/services
- seller can receive notification of appointments or purchases
- seller can set their schedules for both service/product providers
- seller can set open/closed button
- seller can update stock
- seller can receive payments and receipts

Before a student can utilize the app they need to provide information of what they will be offering to avoid selling alcohol or drugs

7. Tools & Methods Used

- Requirements gathering and analysis
- Problem statement definition
- Stakeholder mapping
- Stakeholder Interviews: Conducted with sellers (Nail Techs) and buyers.
- Process Mapping: "As-Is" vs "To-Be" journey mapping.
- Prototyping: UI/UX design requirements.
- Technical Stack (Proposed): Python for backend logic, SQL for inventory databases, and Power BI for tracking campus trade trends.

8. Expected Impact

- Reduced spam posting in residence WhatsApp group
- Increased visibility for student businesses beyond their immediate section
- Improved customer experience — easier to find products and services
- Platform for serious student entrepreneurs to grow their businesses
- Academic Balance: Automated scheduling allows student entrepreneurs to "close shop" during exam weeks, reducing academic stress.
- Safety & Privacy: Decreases the need to share personal contact numbers and room details publicly in large groups.
- Economic Formalization: Helps student businesses build a "track record" or digital portfolio that can be used for future entrepreneurship endeavours

TO-BE Process: Student Purchase Journey (SOM App)

Step	Actor	Action	Improvement Over AS-IS
1	Buyer	Browses real-time inventory in the SOM app	Eliminates PMing: No need to ask "is it available?"
2	Buyer	Selects items and adds to digital cart	Efficiency: Order is clear and documented automatically.
3	Buyer	Proceeds to checkout and selects payment	Standardization: No back-and-forth asking for payment methods.
4	Buyer	Pays via integrated wallet/QR code	Removes Manual Entry: Money is held securely or ready for release.
5	Seller	Receives instant order notification and room #	Automation: No manual text coordination required.
6	Seller	Delivers item to the location	Speed: Seller knows exactly where to go.
7	Buyer	Scans Seller's QR code to confirm receipt	Eliminates "Awkward Moment": Payment is released instantly.

9. Outcome

Since the solution is in the design phase, success is defined by the following validation metrics:

- **Transaction Speed:** A projected 60% reduction in checkout time by replacing manual beneficiary setup with integrated QR-code payments.
- **Seller Consistency:** 100% of interviewed service providers (Nail Techs) identified the "Appointment Limit" feature as essential for balancing academics and business.
- **Market Transparency:** The centralized directory eliminates "Information Asymmetry," allowing buyers to find alternatives instantly when a primary seller is "Closed."

10. Lessons Learned

- **Observation is Elicitation:** I learned that an analyst's most powerful tool is Empathy. By acting as a buyer, I identified a core "Pain Point" (the payment delay) that was not explicitly mentioned in initial interviews.
- **The Introvert Advantage:** I discovered that being an introvert allows for deep observation. While I found spontaneous interviews challenging, I succeeded by using Prototype-Led Elicitation, where the design facilitates the conversation.
- **Structural Alignment:** This project taught me that a technical solution must solve a human problem. The "Open/Closed" button isn't just a feature; it's a tool for student mental health and academic success.